

AI Use Declaration Form

Course: Introduction to Artificial Intelligence

Lab Title: Lab 1 Part A — Uninformed Search Algorithms

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GitHub Repository Link: https://github.com/hswee200/ai_Labs.git

Date Submitted: Saturday 20th June

1. AI Use Summary

Question	Student Response
Did you use any AI tool for this lab?	Yes
Estimated percentage of the work influenced by AI	30%
Did you attach evidence of AI use where applicable?	Yes

2. Details of AI Use

Complete the table below for each AI tool used. Add more rows where necessary.

Name of AI Tool Used	Purpose for Using the Tool	Prompt or Instruction Given to the Tool	Part(s) of the Work Influenced by the Tool	How I Verified, Edited, or Improved the AI Output
Chat Gpt	To understand informed search concepts, debug code, and clarify algorithm behaviour	i have gotten confused to simply explain all what we have done up until now explain are goal and hwo we are planing on getting there, why we did it what was its result, how it did it how it can change what were the options what would have happened if we did eg. A instead of B. do this	Lab 1 Part A (BFS, DFS, DLS, IDS, reached list, search trees, reflection questions) BestFirstSearch implementation, Greedy Best-First Search, A*, Weighted A*, heuristic functions, reflection questions, debugging	I cross-checked explanations with lecture notes, the lab handout, and experimental results from my notebook before using them.

		simply but deeply		

3. Attachment of AI Output Evidence

Where required, attach evidence of AI use. Tick all that apply. Add more rows where necessary.

Evidence Type	Attached? (Yes, No, N/A)	File Name
AI-generated draft	No	
Prompt history	Yes	
Screenshot of AI interaction	Yes	
Exported AI conversation	No	
AI-generated code snippet	Yes	
Revised version showing student input	No	

4. Student Declaration

I declare that:

Declaration Statement	Response (Yes/No)
The submitted work is my own work.	Yes
Any use of AI tools has been clearly declared in this form.	Yes
The prompts, instructions, and AI-generated outputs have been disclosed where applicable.	Yes
I reviewed, tested, edited, and improved any AI-generated content before submission.	Yes
My AI usage does not exceed 25% of the entire work.	No
I understand that undeclared or excessive AI use may be treated as academic misconduct.	Yes

i have gotten confused to simply explain all what we have done up until now explain are goal and hwo we are planing on getting there, why we did it what was its result, how it did it how it can change what were the options what would have happened if we did eg. A instead of B. do this simply but deeply

Good. You're at the point where many students get lost because you've learned several search algorithms and heuristic concepts separately, but haven't connected them into one story.

Let's build the entire picture.

The Big Goal

Imagine you're building a drone.

The drone starts here:

Plain text

S

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and needs to reach:

Plain text

G

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The world may contain:

• walls



Ask anything

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Okay, so let me see if I get all this whole. So we basically have an initial state and we have a goal, and the whole point of this is to get to that goal. But now, what we are basically trying to figure out is how are we basically going to get to that goal, or in what way are we going to get the agent to get to that goal? Because the agents, the agents can get to that goal in several ways. But then, what we are basically trying to find out is how exactly, how exactly we can find the best path from the start to the, to the end, right? And we're coming along BFS, and with the best first search, it goes through, um, I would say, I want to look at the map with the nodes as like a normal map, and let's say each node is some sort of a landmark that all kind of lead to the same, that are supposed to be lead to the same goal, but there are just kind of like different directions. So the way the BFS does it is searches through the nodes and what's the name, and levels. So it goes to one level, then goes to another level, and that's how it goes to the nodes. And then we have also the frontier and the wish list. And so the Frontier is basically the list of nodes that are stored, are waiting to be explored by the agents, and we have the wish list, and the wish list is basically the nodes that have been explored, and then we have it to prevent the AI from wasting, you know, wasting resources, wasting time to go back to nodes they've already, you know, visited, and explore those nodes in a FIFO manner. So it always takes the first node out first. And yeah, the first node that is put into the list first, you know. So first of all, basically, at what point is the Frontier list mean? Is the Frontier list the map with the A, like let's say the A mapping to B, mapping to C, and then B mapping to C, mapping to E, is that the Frontier? And the wish list is when it goes through that map and stores the nodes. So when it does that, the, so is the FIFO being applied to the wish list? Is that where the FIFO is being applied to? And when it, and the FIFO is basically the order it takes to

Ask anything

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think that?

You're actually much closer than you think. Your confusion comes from mixing up the **graph/map**, the **frontier**, the **reached list**, and the **heuristic**. Let's separate them.

Part 1: The map itself

Suppose we have:

Plain text

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This is the **search space** (the world/map).

It already exists before the algorithm starts.

This is **NOT** the frontier.

This is **NOT** the reached list.

It is simply the world the agent is navigating.

Part 2: What is the Frontier?

The frontier is the list of nodes that have been discovered ↓ but not yet explored.

Ask anything



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Now let's jump to heuristics

You said:

The heuristic is basically an estimate of how far I am from the goal.

Exactly.

That is 100% correct.

Suppose:

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You are in Accra

Goal = Kumasi

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Somebody asks:

Plain text

How far is Kumasi?

Copy

You estimate:

Plain text

250 km

Copy

That estimate is the heuristic.



Ask anything



Auto





This screenshot actually answers your biggest confusion.

Look at this sentence on the slide:

"The direct route through the cost-3 band costs 12, while the calm detour costs 10. An overconfident heuristic charges straight ahead."

Let's draw what is happening.

The Map

Start:

Plain text
S

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Goal:

Plain text
G

Copy



Ask anything

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Student Signature: VAC

Date: 20 June 2026